



FACULTY OF BUSINESS & TECHNOLOGY

BACHELOR IN INFORMATION TECHNOLOGY

BIT4543 - ARTIFICIAL INTELLIGENCE

Project Title: Student Performance Prediction System

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DECLARATION

We declare that this project report entitled “Student Performance Prediction System” is the original work of the members of Group 9. The work presented in this report has been prepared for the BIT 4543 Artificial Intelligence group project.

All sources of information, datasets, research materials and published works used in this project have been acknowledged and referenced appropriately.

ACKNOWLEDGEMENT

We would like to express our sincere appreciation to our lecturer for the guidance and knowledge provided throughout the BIT 4543 Artificial Intelligence course.

We would also like to thank all group members for their contributions to the development of the Student Performance Prediction System. The completion of this project required collaboration in dataset preparation, machine learning development, knowledge base preparation, AI integration, application development, testing and documentation.

Finally, we acknowledge the creators of the Open University Learning Analytics Dataset (OULAD), which provided the academic data used in the development and evaluation of this project.

EXECUTIVE SUMMARY

Educational institutions generate large amounts of academic and learning-related data, but identifying students who may experience academic difficulties remains challenging. In many cases, students are identified as struggling only after formal assessments or examinations have already taken place. This can delay appropriate academic intervention.

This project presents a Student Performance Prediction System that applies Artificial Intelligence and Machine Learning techniques to analyse student academic information and support the identification of students who may require academic attention. The system uses the Open University Learning Analytics Dataset (OULAD),

which contains anonymised information about students, courses, assessments and interactions with the Virtual Learning Environment.

A Random Forest Classifier was implemented as the main machine learning model. The implemented model uses ten features from studentInfo.csv, consisting of eight categorical features (code module, code presentation, gender, region, highest education, IMD band, age band and disability) and two numerical features (number of previous attempts and studied credits) to predict student academic outcomes.

In addition to the machine learning component, the project implements an academic intervention workflow. A structured knowledge base identifies different student risk levels and academic risk factors such as low GPA, poor attendance, missed assignments and low participation. The workflow retrieves relevant guidance and integrates it with a Generative AI component to produce concise and supportive academic intervention recommendations.

The developed system demonstrates how Machine Learning, rule-based reasoning, knowledge representation and Generative AI can be integrated into a single application to support student performance monitoring and academic intervention.

CHAPTER 1: INTRODUCTION

1.1 Background

The education sector increasingly relies on digital technologies to manage teaching, learning and student information. Learning management systems, online learning platforms and institutional databases collect significant amounts of information relating to student activities and academic performance. This information can potentially be analysed to identify learning patterns and predict future academic outcomes.

Artificial Intelligence (AI) provides techniques that can process large amounts of data and identify patterns that may not be immediately visible through manual analysis. Machine Learning, which is a branch of AI, enables computer systems to learn patterns from historical data and make predictions based on new data.

Student performance prediction is an important application of Machine Learning in education. By analysing academic information such as assessment performance, course information and learning engagement, predictive models may assist institutions in identifying students who could require additional academic support.

The Student Performance Prediction System was developed to explore the use of Artificial Intelligence for analysing student academic information. The project uses the Open University Learning Analytics Dataset (OULAD), which contains anonymised information relating to students and their learning activities.

The system combines several AI-related components. A Machine Learning model is used to predict academic outcomes based on selected student features. A knowledge base and rule-based workflow are used to identify academic risk factors and retrieve suitable intervention guidance. A Generative AI component is then used to generate concise and supportive recommendations.

The project therefore demonstrates an integrated approach to student performance monitoring by combining predictive analytics and AI-generated academic guidance.

1.2 Problem Statement

Educational institutions may lack an automated and data-driven mechanism for continuously analysing student academic progress. Faculty members and academic advisors may depend on manual reviews, assessment results or periodic academic reports to identify students who are experiencing difficulties.

This creates several challenges.

1.2.1 Delayed Risk Identification

Traditional academic monitoring methods may depend heavily on examination results or manual review processes. As a result, students experiencing academic difficulties or disengagement may not be identified early enough for effective intervention.

1.2.2 Lack of Predictive Insights

Without predictive analysis, institutions may find it difficult to estimate future academic outcomes using historical and current student information. This limits the ability to identify patterns associated with different academic outcomes.

1.2.3 Ineffective Intervention Alignment

Academic advisors may have limited time and resources. Without structured information about student risk factors, it may be difficult to prioritise students and provide appropriate recommendations.

Therefore, there is a need for a system that can analyse student academic information, apply Machine Learning techniques to predict outcomes and provide structured academic intervention guidance.

1.3 Project Objectives

The objectives of this project are:

1. To develop an Artificial Intelligence-based Student Performance Prediction System for analysing student academic information.
2. To prepare and process relevant student data from the Open University Learning Analytics Dataset.
3. To develop and evaluate a Random Forest Machine Learning model for predicting student academic outcomes.
4. To develop a knowledge base and rule-based workflow for identifying academic risk factors and intervention priorities.
5. To integrate a Generative AI component to generate concise, professional and supportive academic intervention recommendations.
6. To develop a user interface that allows users to interact with the student performance detection workflow.

1.4 Scope of Project

The scope of the Student Performance Prediction System includes the following components.

1.4.1 Dataset

The project uses the Open University Learning Analytics Dataset (OULAD). Relevant information is obtained from OULAD files relating to student information, assessments, registrations and learning activities.

1.4.2 Machine Learning

The project implements a Random Forest Classifier using selected academic features:

- Average assessment score
- Assessment count
- Studied credits
- Number of previous attempts

The model predicts student academic outcome categories based on the available training data.

1.4.3 Academic Intervention Workflow

The system includes a structured knowledge base containing:

- Low-risk guidance
- Medium-risk guidance
- High-risk guidance
- Risk-factor-specific recommendations

The workflow identifies academic risk factors using predefined rules.

1.4.4 Generative AI

The project integrates a Generative AI workflow using the Gemini API to generate concise academic intervention recommendations based on structured student information and retrieved knowledge-base guidance.

1.4.5 Application Development

A prototype application was developed to demonstrate the integration of the machine learning and AI workflow components.

1.4.6 Limitations of Scope

The project is an academic prototype and is not intended to replace professional academic advisors or institutional decision-making systems.

The system also depends on the structure and quality of the OULAD dataset and does not contain real-time data from the user's university.

1.5 Significance of the Project

The Student Performance Prediction System is significant because it demonstrates how Artificial Intelligence can support educational data analysis.

Benefits to Students

Students may benefit from earlier identification of academic difficulties and more structured intervention recommendations.

Benefits to Academic Advisors

Academic advisors may use predictive and structured information to support monitoring and prioritisation.

Benefits to Educational Institutions

The project demonstrates the potential of educational data analytics to support evidence-based decision-making.

Benefits to Future Research

The project provides a foundation for future improvements involving additional datasets, improved features, advanced Machine Learning algorithms and real-time institutional data.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview of Artificial Intelligence Technologies

2.1.1 Artificial Intelligence

Artificial Intelligence refers to computer systems designed to perform tasks that normally require aspects of human intelligence, such as learning, reasoning, pattern recognition and decision-making.

AI has been applied in many areas, including healthcare, finance, transportation and education. In education, AI can support personalised learning, automated assessment, learning analytics and student performance prediction.

The Student Performance Prediction System applies AI through the integration of Machine Learning, rule-based reasoning and Generative AI.

2.1.2 Machine Learning

Machine Learning is a branch of Artificial Intelligence that enables computer systems to learn patterns from data.

In supervised learning, a model is trained using historical examples containing input variables and known output categories. After training, the model can be used to predict outcomes for new data.

For student performance prediction, Machine Learning algorithms can analyse variables such as:

- Assessment results
- Course information
- Previous academic attempts
- Learning engagement
- Registration information

The model developed in this project uses supervised classification.

2.1.3 Random Forest

Random Forest is an ensemble Machine Learning algorithm that combines multiple decision trees.

Instead of relying on a single decision tree, Random Forest creates multiple trees and combines their predictions. This approach can improve robustness and reduce the risk of relying excessively on one individual decision tree.

Random Forest is suitable for classification tasks involving structured tabular data. It can also provide feature importance information, allowing researchers to examine which variables contribute to predictions.

The Student Performance Prediction System uses the RandomForestClassifier implementation from the Scikit-learn library.

2.1.4 Generative Artificial Intelligence

Generative AI refers to AI systems capable of generating new content, including text.

Large Language Models can process structured prompts and generate human-readable responses. In educational applications, Generative AI may assist in generating explanations, recommendations and personalised guidance.

However, Generative AI output should be carefully controlled. In this project, the AI component is guided using:

- A fixed system prompt
- Structured student information
- Few-shot examples
- Knowledge-base information
- Risk-factor-specific guidance

This approach is intended to produce recommendations that are relevant, concise and professional.

2.1.5 Natural Language Processing

Natural Language Processing (NLP) is a field of AI concerned with enabling computers to process and generate human language.

Although the main predictive model in this project is based on structured tabular data, NLP-related capabilities are used through the Generative AI component. The system converts structured student information into a prompt and receives a natural-language intervention recommendation.

2.2 Related Works

Educational data mining and learning analytics research has investigated the use of student data to understand learning behaviour and predict academic outcomes.

The OULAD dataset was introduced as a dataset designed to support research in learning analytics. It contains anonymised information relating to students, courses, assessments and Virtual Learning Environment interactions.

Machine Learning research commonly applies classification algorithms to predict student outcomes. Different studies have explored Decision Trees, Random Forest, Support Vector Machines and other classification methods.

Random Forest is frequently considered suitable for structured datasets because it can model complex relationships between variables and provide feature importance values.

However, prediction alone may not be sufficient for academic support. An educational system should also provide understandable and actionable information that can support intervention.

The following table summarises the relationship between previous approaches and the present project.

Study/System	Method Used	Strengths	Limitations
Educational data mining systems	Machine Learning classification	Can identify patterns in academic data	May provide predictions without intervention guidance
Learning analytics systems	Student activity analysis	Supports understanding of learning behaviour	May require complex institutional data
OULAD-based research	Predictive analytics	Uses a well-documented educational dataset	Results may depend on selected features and methodology
Generative AI educational assistants	Large Language Models	Can generate human-readable guidance	Responses may require structured control
Present project	Random Forest + Knowledge Base + Generative AI	Combines prediction, risk analysis and recommendations	Prototype limitations and moderate model performance

2.3 Research Gap

Although Machine Learning can be used to predict student outcomes, a prediction by itself may not provide sufficient information about what action should be taken.

For example, predicting that a student may experience academic difficulty does not automatically explain:

- Which factors require attention
- What intervention should be prioritised
- How recommendations should be communicated

Another challenge is that Generative AI systems can generate useful recommendations but may not automatically follow specific institutional guidance unless relevant context is provided.

The Student Performance Prediction System addresses this gap by integrating:

- Machine Learning-based academic outcome prediction.
- Rule-based identification of academic risk factors.
- A structured academic intervention knowledge base.
- Generative AI-based recommendation generation.

The objective is to move beyond prediction alone by providing structured and action-oriented academic guidance.

2.4 Conceptual Framework

The conceptual framework of the system consists of three main stages.

Input

The system receives student-related academic information, including:

- Assessment information
- Assessment scores
- Studied credits
- Previous attempts
- Risk profile information for the intervention workflow

Process

The processing stage includes:

- Dataset preparation
- Data preprocessing
- Feature engineering
- Random Forest prediction
- Risk-factor identification

- Knowledge-base retrieval
- Prompt construction
- Generative AI processing

Output

The system produces:

- Predicted academic outcome
- Identified academic risk factors
- Risk-level guidance
- Academic intervention recommendation

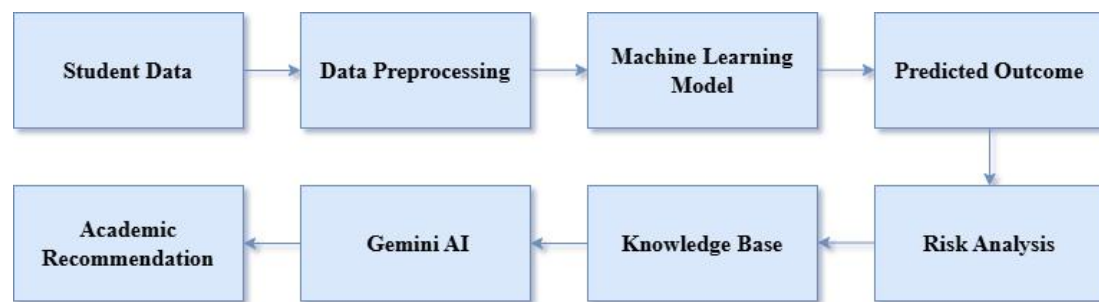


Figure 2.1: Conceptual framework diagram

CHAPTER 3: METHODOLOGY

3.1 Development Methodology

The project followed an iterative development approach.

The development process involved several stages:

1. Problem identification
2. Dataset collection
3. Data preparation
4. Machine Learning model development
5. Knowledge base preparation
6. API integration
7. AI workflow integration

8. Application development
9. Testing and evaluation
10. Documentation and finalisation

The iterative approach allowed the project components to be developed and tested progressively.

3.2 System Architecture

The OULAD dataset provides the academic and student information used in the project. For the final Random Forest implementation, the model training process uses the studentInfo.csv file. Other OULAD files are available for future feature expansion.

Dataset Layer

The OULAD dataset provides the academic data used for Machine Learning development.

Data Processing Layer

Relevant CSV files are loaded and processed. Student and assessment information is combined to create features suitable for the model.

Machine Learning Layer

The Random Forest Classifier is trained using the processed dataset and used to predict student academic outcomes.

Knowledge Layer

A JSON-based academic intervention knowledge base stores risk-level guidance and risk-factor-specific recommendations.

AI Workflow Layer

The workflow combines:

- Student information

- Identified risk factors
- Knowledge-base information
- Prompt engineering
- Few-shot examples

The integrated prompt is sent to the Gemini model.

Application Layer

The application provides an interface through which users can interact with the system and view the generated results.

3.3 Data Collection

3.3.1 Dataset Source

The dataset used in this project is the Open University Learning Analytics Dataset (OULAD).

OULAD is a higher-education dataset developed to support learning analytics research. The dataset contains anonymised information about students, courses, assessments and interactions with a Virtual Learning Environment.

According to Kuzilek et al. (2017), the dataset contains information relating to 32,593 students across 22 course presentations.

3.3.2 Dataset Files

The main OULAD files include the following.

File	Description
studentInfo.csv	Student demographic and final outcome information
studentRegistration.csv	Student registration information
courses.csv	Course and module information
assessments.csv	Assessment information
studentAssessment.csv	Student assessment results

vle.csv	Virtual Learning Environment resource information
studentVle.csv	Student interaction with VLE resources

These files can be connected using relevant identifiers such as:

- Student identifier
- Course/module identifier
- Presentation identifier
- Assessment identifier

3.3.3 Dataset Suitability

OULAD was selected because the project focuses on university-level student performance.

The dataset contains information related to:

- Student demographics
- Course information
- Student registration
- Assessments
- Assessment results
- Virtual Learning Environment activities

The availability of multiple types of educational information makes the dataset suitable for Machine Learning experimentation.

	A	B	C	D	E	F	G	H	I	J	K	L
1	code_module	code_presentation	id_student	gender	region	highest_education	imd_band	age_band	num_of_prev_attempts	studied_credits	disability	final_result
2	AAA	2013J	11391	M	East Anglian Region	HE Qualification	90-100%	55<=	0	240	N	Pass
3	AAA	2013J	28400	F	Scotland	HE Qualification	20-30%	35-55	0	60	N	Pass
4	AAA	2013J	30268	F	North Western Region	A Level or Equivalent	30-40%	35-55	0	60	Y	Withdrawn
5	AAA	2013J	31604	F	South East Region	A Level or Equivalent	50-60%	35-55	0	60	N	Pass
6	AAA	2013J	32885	F	West Midlands Region	Lower Than A Level	50-60%	0-35	0	60	N	Pass
7	AAA	2013J	38053	M	Wales	A Level or Equivalent	80-90%	35-55	0	60	N	Pass
8	AAA	2013J	45462	M	Scotland	HE Qualification	30-40%	0-35	0	60	N	Pass
9	AAA	2013J	45642	F	North Western Region	A Level or Equivalent	90-100%	0-35	0	120	N	Pass
10	AAA	2013J	52130	F	East Anglian Region	A Level or Equivalent	70-80%	0-35	0	90	N	Pass
11	AAA	2013J	53025	M	North Region	Post Graduate Qualification	?	55<=	0	60	N	Pass
12	AAA	2013J	57506	M	South Region	Lower Than A Level	70-80%	35-55	0	60	N	Pass
13	AAA	2013J	58873	F	East Anglian Region	A Level or Equivalent	20-30%	0-35	0	60	N	Pass
14	AAA	2013J	59185	M	East Anglian Region	Lower Than A Level	60-70%	35-55	0	60	N	Pass
15	AAA	2013J	62155	F	North Western Region	HE Qualification	50-60%	0-35	0	60	N	Pass
16	AAA	2013J	63400	M	Scotland	Lower Than A Level	40-50%	35-55	0	60	N	Pass
17	AAA	2013J	65002	F	East Anglian Region	A Level or Equivalent	70-80%	0-35	0	60	N	Withdrawn
18	AAA	2013J	70464	F	West Midlands Region	A Level or Equivalent	60-70%	35-55	0	60	N	Pass
19	AAA	2013J	71361	M	Ireland	HE Qualification	?	35-55	0	60	N	Pass
20	AAA	2013J	74372	M	East Anglian Region	A Level or Equivalent	20-Oct	35-55	0	150	N	Fail
21	AAA	2013J	75091	M	South Western Region	A Level or Equivalent	30-40%	35-55	0	60	N	Pass
22	AAA	2013J	77367	M	East Midlands Region	A Level or Equivalent	30-40%	0-35	0	60	N	Pass
23	AAA	2013J	91265	M	North Western Region	HE Qualification	0-10%	0-35	0	60	N	Pass
24	AAA	2013J	94961	M	South Region	Lower Than A Level	70-80%	35-55	0	60	N	Withdrawn
25	AAA	2013J	98094	M	Wales	Lower Than A Level	70-80%	35-55	0	60	N	Pass

Figure 3.2: OULAD dataset

3.4 Data Preprocessing

Data preprocessing was required before training the Machine Learning model.

The major preprocessing activities included:

3.4.1 Loading Data

The final Machine Learning model was trained using studentInfo.csv. Other OULAD files, including studentAssessment.csv, assessments.csv, studentRegistration.csv, vle.csv and studentVle.csv, were examined as potential sources for future feature expansion.

3.4.2 Data Inspection

The dataset structure was examined by checking:

- Column names
- Data types
- Sample records
- Dataset dimensions

This step was important for understanding the available information and selecting appropriate variables.

3.4.3 Feature Engineering

Features were created or selected from the processed student data.

The main features used in the implemented Machine Learning model were:

1. Average assessment score
2. Assessment count
3. Studied credits
4. Number of previous attempts

These features represent aspects of academic performance and previous study history.

3.4.4 Handling Data for Model Training

The processed data was prepared for supervised classification.

The target variable used in the actual implemented training workflow was the student's final academic outcome.

The classes included:

- Distinction
- Fail
- Pass
- Withdrawn

The final processed dataset used for model training contained 26,074 students.

3.5 AI Model Development

3.5.1 Algorithm Selection

A Random Forest Classifier was selected as the main Machine Learning algorithm.

Random Forest was chosen because:

- It is suitable for classification tasks.

- It works well with structured tabular data.
- It can model non-linear relationships.
- It combines multiple decision trees.
- It can provide feature importance values.

3.5.2 Model Configuration

The project implementation used the Random Forest algorithm from Scikit-learn.

The previously prepared Machine Learning workflow included the use of:

- Random Forest classification
- Multiple decision trees
- Reproducible model configuration
- Class balancing where configured
- Feature importance analysis

The trained model was saved as:

risk_model.pkl

This allows the trained model to be loaded and reused by the application.

3.5.3 Training and Testing Split

The dataset was divided into:

- 80% training data
- 20% testing data

A stratified train-test split was used to maintain the distribution of outcome categories as appropriately as possible between the training and testing datasets.

The final execution showed:

- Training dataset: 26,074 students
- Testing dataset: 6,519 students

3.5.4 Training Process

The Machine Learning workflow involved the following steps:

1. Load the processed student data.
2. Select the input features.
3. Select the target outcome variable.
4. Divide the data into training and testing sets.
5. Train the Random Forest model.
6. Generate predictions for the testing dataset.
7. Evaluate the predictions.
8. Save the trained model.

3.6 Knowledge Base Preparation

A structured academic intervention knowledge base was prepared to support the AI workflow

```
112 # =====
113 # CREATE ACADEMIC INTERVENTION KNOWLEDGE BASE
114 # =====
115
116 knowledge_base = {
117
118     "knowledge_base_name":
119         "Student Academic Intervention Knowledge Base",
120
121     "version":
122         "1.0",
123
124     "purpose":
125         "Provide academic intervention guidance for students "
126         "identified as being at risk by the machine learning model.",
127
128     # -----
129     # RISK LEVEL GUIDANCE
130     # -----
131
132     "risk_levels": {
133
134         "Low": {
135
136             "description":
137                 "Student performance and engagement indicators "
```

Figure 3.4: Knowledge Base implementation code

3.6.1 Knowledge Base Components

The knowledge base contains:

1. Student academic intervention guidance.
2. Three student risk levels:
 - Low
 - Medium
 - High
3. Four major risk factors:
 - Low GPA
 - Poor attendance
 - Missed assignments
 - Low participation
4. Risk-level-specific intervention recommendations.
5. Risk-factor-specific academic guidance.
6. Seven intervention rules linking conditions to recommended actions.
7. Professional and supportive advisor guidelines.

The knowledge base is stored in JSON format.

```
knowledge_base.json
1 {
2   "knowledge_base_name": "Student Academic Intervention Knowledge Base",
3   "version": "1.0",
4   "purpose": "Provide academic intervention guidance for students identified as being at risk by the machine learning model.",
5   "risk_levels": {
6     "Low": {
7       "description": "Student performance and engagement indicators are generally stable.",
8       "priority": "Routine monitoring",
9       "recommended_actions": [
10        "Continue normal academic monitoring.",
11        "Encourage the student to maintain current study habits.",
12        "Provide positive academic feedback when appropriate."
13      ]
14    },
15    "Medium": {
16      "description": "Student shows one or more academic or engagement concerns that may affect future performance.",
17      "priority": "Early intervention",
18      "recommended_actions": [
19        "Send a supportive check-in message.",
```

Figure 3.5: JSON-based academic intervention knowledge base


```

285 # INTERVENTION RULES
286 # -----
287
288 "intervention_rules": [
289     {
290         "condition":
291             "High risk with multiple risk factors",
292         "priority":
293             "Immediate",
294         "action":
295             "Arrange an academic advisor meeting and "
296             "recommend tutoring or academic support."
297     },
298     {
299         "condition":
300             "High risk with poor attendance",
301         "priority":
302             "Immediate",
303     }
304 ]

```

Figure 3.6: Risk levels and intervention rules

3.6.2 Student Risk Profile

Student information used by the intervention workflow is structured using:

- Risk level
- GPA
- Attendance percentage
- Missed assignments
- Participation score

A Python dataclass was used to represent the student risk profile.

3.7 AI Workflow Integration

The complete AI workflow integrates the following stages:

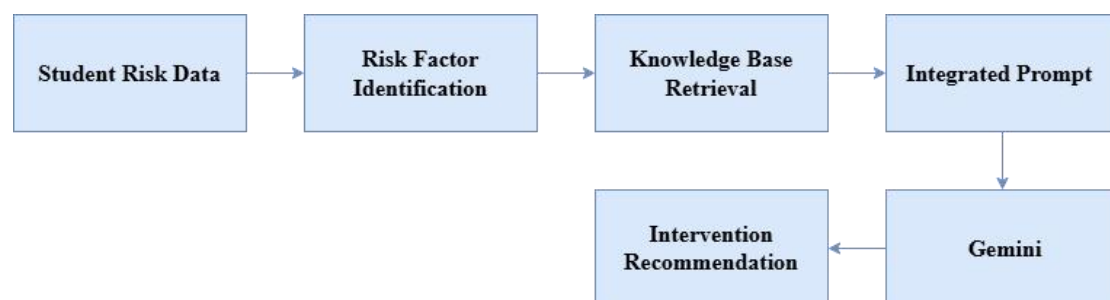


Figure 3.3: AI workflow integration

The workflow performs the following tasks:

1. Loads the prepared knowledge base.
2. Accepts a student risk profile.
3. Identifies risk factors automatically.
4. Retrieves appropriate risk-level guidance.
5. Retrieves risk-factor-specific guidance.
6. Builds an integrated knowledge context.
7. Combines student information and guidance.
8. Sends the information to Gemini.
9. Generates an academic intervention recommendation.

3.7.1 Risk-Factor Rules

The implemented workflow uses predefined thresholds.

Risk Factor	Rule
Low GPA	GPA < 2.0
Poor Attendance	Attendance < 75%
Missed Assignments	3 or more
Low Participation	Participation < 5/10

These rules allow the workflow to identify relevant academic concerns.

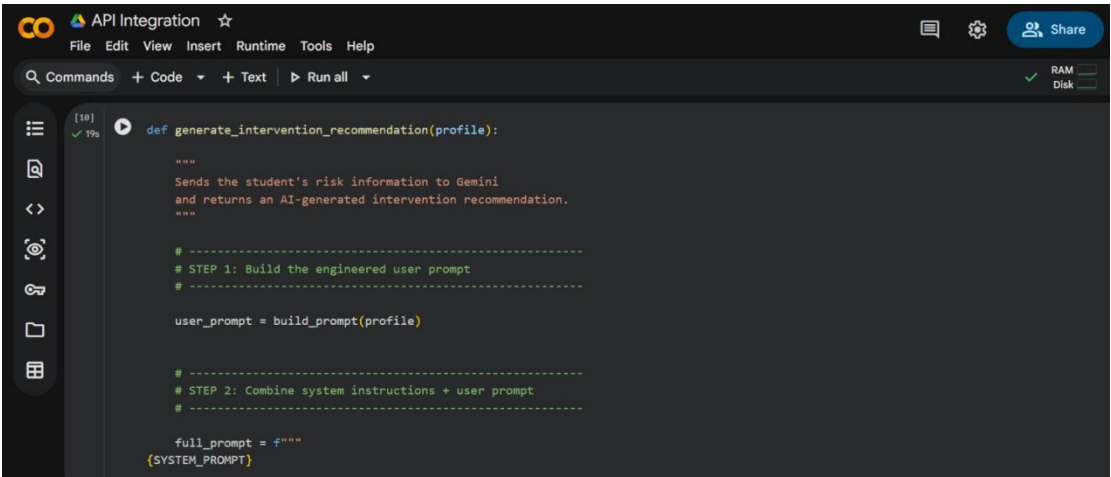
3.8 API Integration

The Generative AI component includes the following features:

1. A fixed system prompt defining the AI as a university academic advisor assistant.
2. Few-shot examples for Low, Medium and High-risk students.
3. Dynamic formatting of student information.
4. AI-generated intervention recommendations.
5. Recommendations restricted to approximately two to three sentences.
6. Professional, supportive and concise output.

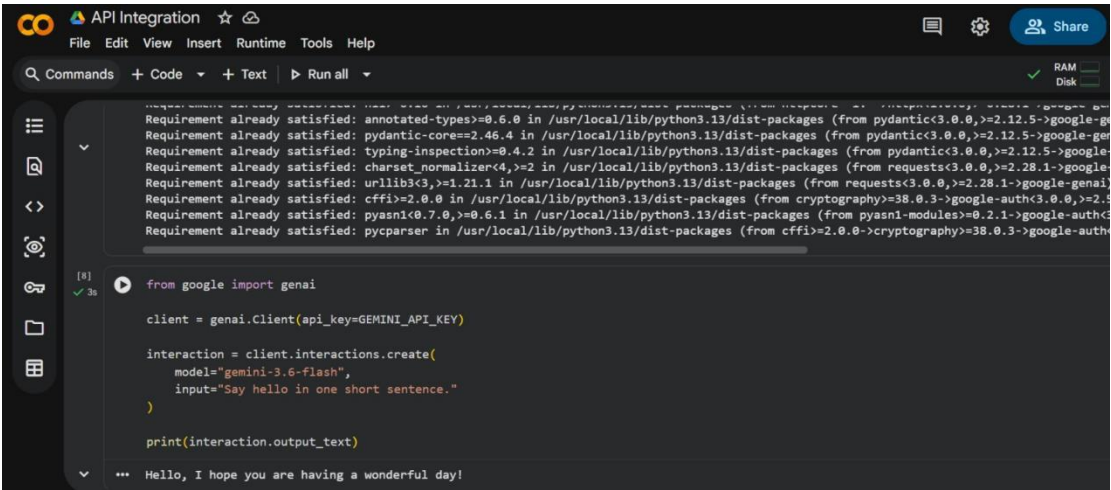
- 7. Recommendations based on identified risk factors.
- 8. Gemini API integration.
- 9. Extraction of the generated response.

The API workflow combines the system prompt, student information, few-shot examples and retrieved knowledge context before sending the final prompt to the Generative AI model.



```
def generate_intervention_recommendation(profile):  
    """  
    Sends the student's risk information to Gemini  
    and returns an AI-generated intervention recommendation.  
    """  
  
    # -----  
    # STEP 1: Build the engineered user prompt  
    # -----  
  
    user_prompt = build_prompt(profile)  
  
    # -----  
    # STEP 2: Combine system instructions + user prompt  
    # -----  
  
    full_prompt = f"""  
    {SYSTEM_PROMPT}
```

Figure 3.9: Generative AI API integration code



```
Requirement already satisfied: annotated-types==0.6.0 in /usr/local/lib/python3.13/dist-packages (from pydantic<3.0.0,>=2.12.5->google-g  
Requirement already satisfied: pydantic-core==2.46.4 in /usr/local/lib/python3.13/dist-packages (from pydantic<3.0.0,>=2.12.5->google-g  
Requirement already satisfied: typing-inspection==0.4.2 in /usr/local/lib/python3.13/dist-packages (from pydantic<3.0.0,>=2.12.5->google-g  
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests<3.0.0,>=2.28.1->google  
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests<3.0.0,>=2.28.1->google-gemai  
Requirement already satisfied: cffi>=2.0.0 in /usr/local/lib/python3.13/dist-packages (from cryptography>=38.0.3->google-auth<3.0.0,>=2.5  
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /usr/local/lib/python3.13/dist-packages (from pyasn1-modules>=0.2.1->google-auth<  
Requirement already satisfied: pycparser in /usr/local/lib/python3.13/dist-packages (from cffi>=2.0.0->cryptography>=38.0.3->google-auth<  
  
[8] ▶ from google import genai  
✓ 3s  
  
client = genai.Client(api_key=GEMINI_API_KEY)  
  
interaction = client.interactions.create(  
    model="gemini-3.6-flash",  
    input="Say hello in one short sentence."  
)  
  
print(interaction.output_text)  
  
... Hello, I hope you are having a wonderful day!
```

Figure 3.10: Gemini API configuration

3.9 Tools and Technologies

Component	Technology
-----------	------------

Programming Language	Python
Data Processing	Pandas
Machine Learning	Scikit-learn
Model	Random Forest Classifier
Model Storage	Joblib / Pickle
Knowledge Base	JSON
Generative AI	Google Gemini API
AI SDK	Google GenAI SDK
Development Environment	Google Colab / Python
Application Prototype	Flask + HTML/CSS/JavaScript
Version Control	GitHub
Dataset	OULAD

CHAPTER 4: SYSTEM DESIGN AND IMPLEMENTATION

4.1 System Requirements

4.1.1 Functional Requirements

The system should be able to:

1. Load student-related data.
2. Process academic information.
3. Generate Machine Learning predictions.
4. Identify relevant student risk factors.
5. Retrieve knowledge-base guidance.
6. Generate AI-based intervention recommendations.
7. Display student-related results through the application interface.
8. Handle invalid or missing input appropriately.

4.1.2 Non-Functional Requirements

Performance

The system should process user input and generate results within a reasonable time.

Usability

The application interface should be simple and understandable.

Reliability

The system should provide validation for invalid inputs where applicable.

Maintainability

The project uses separate Python modules to improve organisation and maintainability.

Security

API keys and sensitive credentials should not be publicly exposed in source code repositories.

4.2 Use Case Design

The main user of the system is an academic user or system operator.

The user can:

- Enter student information.
- Request student performance analysis.
- View predicted information.
- View identified risk factors.
- Receive academic intervention recommendations.

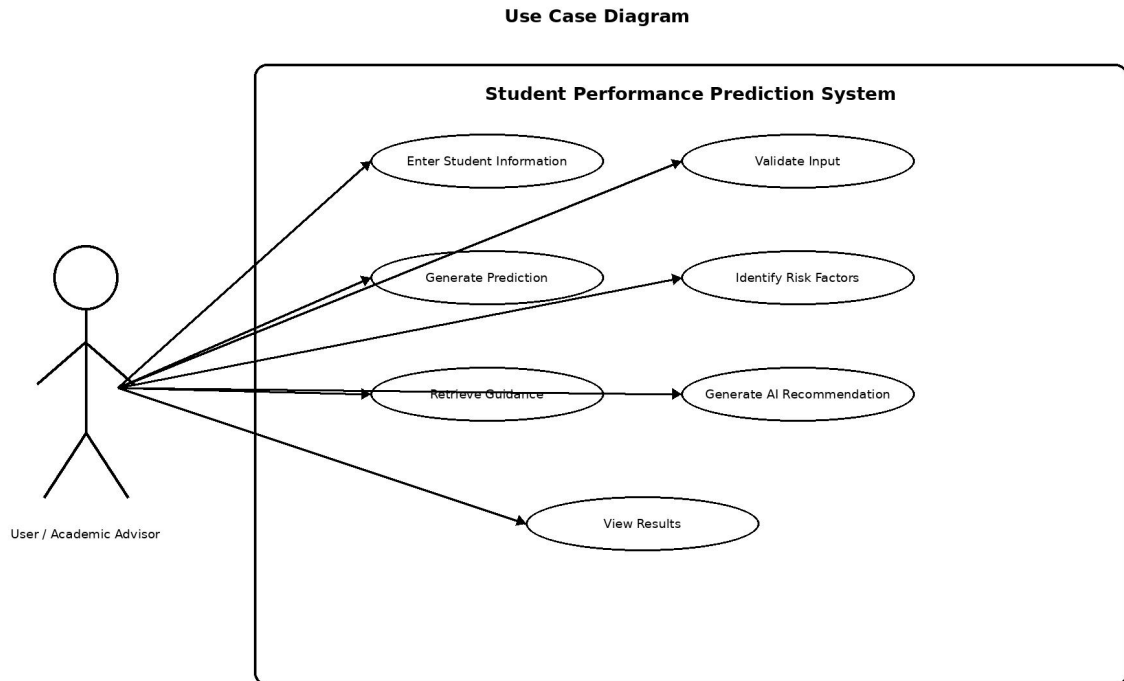


Figure 4.1: Use case diagram

Suggested actors and use cases:

Actor: User

Use Cases:

- Enter Student Information
- Validate Input
- Generate Prediction
- Identify Risk Factors
- Retrieve Guidance
- Generate AI Recommendation
- View Results

4.3 Activity Diagram

The system workflow can be described as follows:

1. User opens the application.

2. User enters the required student information.
3. System validates the input.
4. System processes the information.
5. The Machine Learning component generates the relevant prediction.
6. The risk workflow identifies academic risk factors.
7. The knowledge base retrieves appropriate guidance.
8. Gemini generates an intervention recommendation.
9. The results are displayed to the user.

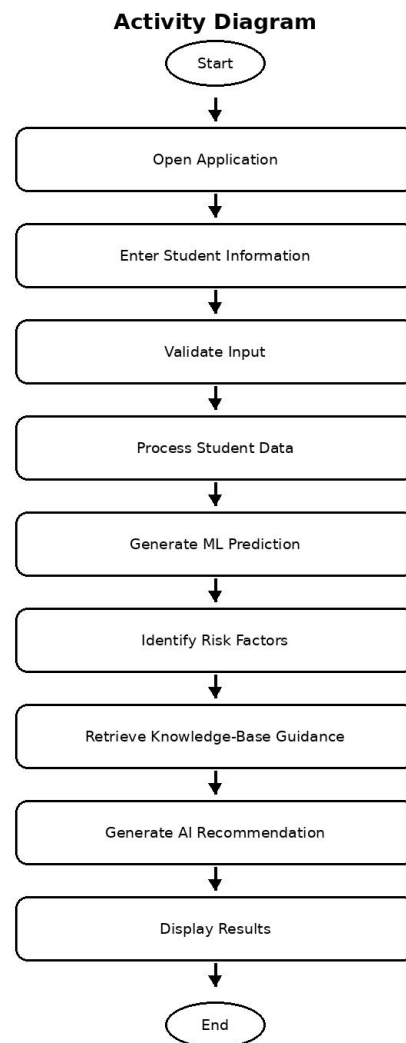


Figure 4.2: Activity diagram

4.4 Data Design

The project does not require a traditional relational database for the prototype.

Instead, the system uses:

- CSV files for dataset storage and processing.
- A .pkl file for the trained Machine Learning model.
- JSON for the academic intervention knowledge base.

The main data flow is:

OULAD CSV Files → Data Processing → Machine Learning Model

and

Student Risk Profile → JSON Knowledge Base → AI Workflow → Recommendation

4.5 User Interface Design

The prototype application was developed to provide a simple interface for interacting with the system.

The interface supports the entry of student information and displays the generated results.

Input validation was implemented to handle situations such as:

- Invalid student IDs
- Non-numeric input
- Missing or empty input

The interface also displays relevant output such as:

- Student information
- Prediction results
- Summary information
- Academic intervention recommendations

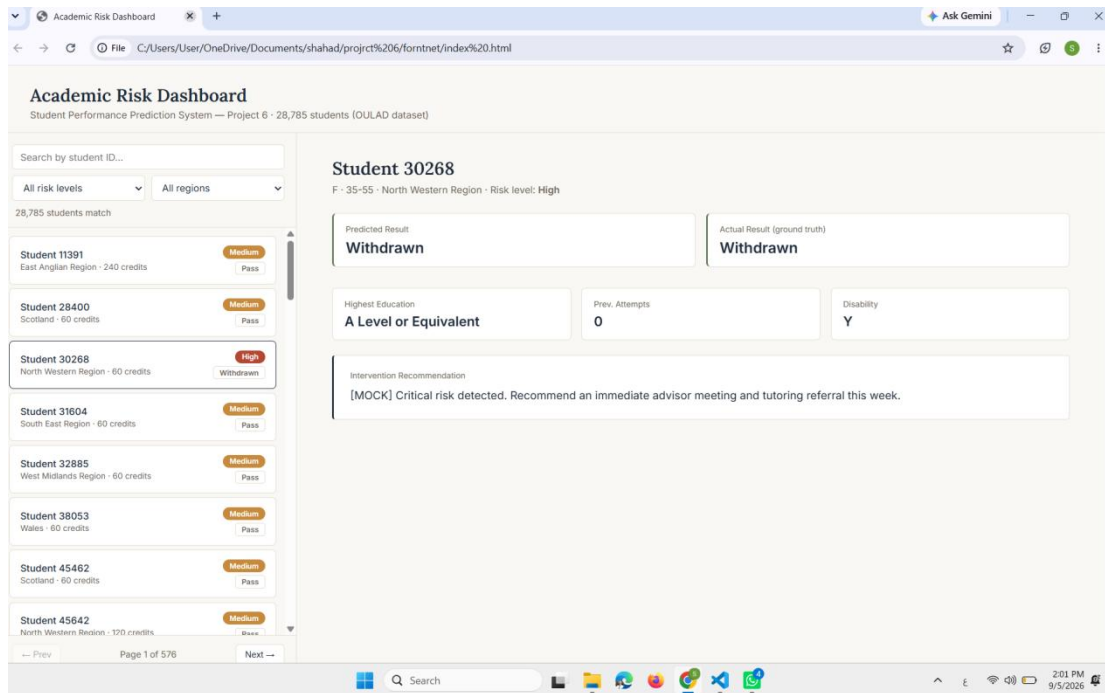


Figure 4.3: Application interface

4.6 System Implementation

The project was organised into several Python modules.

4.6.1 Machine Learning Module

The Machine Learning implementation is contained in files related to:

- Model training
- Performance evaluation
- Model storage

The trained model is saved as:

risk_model.pkl

4.6.2 Model Module

The model-related module manages the loading and use of the trained model.

4.6.3 Knowledge Base Module

The knowledge base preparation module creates and manages the structured academic intervention information.

It includes:

- Risk levels
- Risk factors
- Intervention rules
- Academic guidance

4.6.4 Prompt Engineering Module

The prompt engineering component prepares structured prompts for the Generative AI model.

It includes:

- System instructions
- Student information
- Few-shot examples
- Knowledge-base context

4.6.5 API Integration Module

The API integration module communicates with the Gemini API and processes the generated response.

4.6.6 AI Workflow Integration Module

The AI workflow integration module connects:

- Student risk information
- Risk-factor identification
- Knowledge-base retrieval
- Prompt engineering
- Generative AI output

This module performs the end-to-end intervention workflow.

4.6.7 Application Module

The application module provides the user-facing interface.

The prototype was tested to ensure that the main input and output workflow operated correctly.

CHAPTER 5: TESTING AND EVALUATION

5.1 Testing Strategy

The project testing involved:

1. Functional testing
2. Input validation testing
3. Workflow integration testing
4. Machine Learning model evaluation

The implemented testing focused mainly on validating the functionality of the prototype and the end-to-end AI workflow.

5.2 Test Cases

Table 5.1: Functional Test Cases

Test Case	Expected Result	Actual Result	Status
Valid student information	System processes the information	Information processed successfully	Pass
Invalid student ID	System displays an error	Invalid ID message displayed	Pass
Non-numeric ID input	System rejects invalid input	Validation message displayed	Pass
Empty input	System does not generate invalid	Input remains unchanged or	Pass

	results	requires valid data	
Model loading	Saved model loads successfully	Model available for prediction	Pass
Knowledge-base loading	JSON knowledge base loads	Guidance retrieved successfully	Pass
Low-risk profile	Appropriate guidance generated	Workflow completed	Pass
Medium-risk profile	Appropriate guidance generated	Workflow completed	Pass
High-risk profile	Multiple risk factors identified	Workflow completed	Pass

5.3 AI Workflow Validation

The intervention workflow was tested using three student profiles.

Table 5.2: Test Student Profiles

Test Case	Risk Level	GPA	Attendance	Missed Assignments	Participation
TEST-LOW	Low	3.4	91%	0	8/10
TEST-MEDIUM	Medium	2.6	78%	2	5/10
TEST-HIGH	High	1.9	60%	4	3/10

5.3.1 Low-Risk Test

The Low-risk profile represents a student with relatively strong academic indicators.

The system processes the student profile and retrieves guidance associated with routine monitoring.

Expected behaviour:

- No major risk factors should dominate the recommendation.
- The guidance should encourage continued positive academic behaviour.

5.3.2 Medium-Risk Test

The Medium-risk profile represents a student who may require early academic support.

The system retrieves medium-priority guidance and generates an appropriate recommendation based on the student's information.

5.3.3 High-Risk Test

The High-risk profile contains:

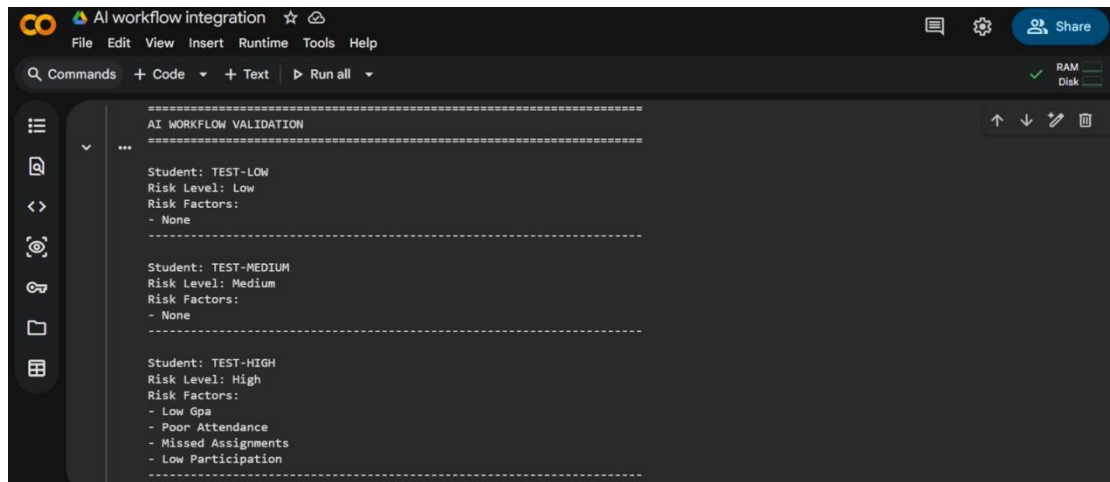
- GPA below the predefined threshold.
- Attendance below 75%.
- Four missed assignments.
- Participation below 5/10.

The system therefore identifies multiple academic risk factors and retrieves high-priority intervention guidance.

Expected risk factors include:

- Low GPA
- Poor attendance
- Multiple missed assignments
- Low participation

The system then generates an academic recommendation that focuses on immediate support and improvement.

The image shows a software interface titled "AI workflow integration" with a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar (Commands, + Code, + Text, Run all). On the left is a sidebar with icons for file management and execution. The main area displays "AI WORKFLOW VALIDATION" results for three test cases, separated by dashed lines. Each case shows a student ID, risk level, and a list of risk factors.

```
=====
AI WORKFLOW VALIDATION
=====

Student: TEST-LOW
Risk Level: Low
Risk Factors:
- None

-----

Student: TEST-MEDIUM
Risk Level: Medium
Risk Factors:
- None

-----

Student: TEST-HIGH
Risk Level: High
Risk Factors:
- Low Gpa
- Poor Attendance
- Missed Assignments
- Low Participation
=====
```

Figure 5.1: Low-risk, medium-risk and high-risk workflow test result

5.4 MODEL EVALUATION

5.4.1 Evaluation Metrics

The Random Forest classification model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Classification report

The model was trained on 26,074 students and evaluated on a testing dataset containing 6,519 students.

5.4.2 Actual Model Performance

The actual terminal execution of the implemented model produced the following results.

```
[12]  # MAKE PREDICTIONS
# =====

y_pred = rf_model.predict(X_test)

print("Predictions generated successfully.")

print("\nFirst 20 predictions:")
print(y_pred[:20])

▼ ... Predictions generated successfully.

First 20 predictions:
['Medium' 'Low' 'Low' 'Low' 'Medium' 'Low' 'High' 'High' 'Medium' 'Low'
 'Low' 'Low' 'Low' 'Low' 'Low' 'High' 'High' 'Low' 'High' 'High']
```

Figure 5.5: Random Forest model training and evaluation output

Table 5.3: Overall Model Performance

Evaluation Metric	Result
Accuracy	44.62%
Weighted Precision	0.43
Weighted Recall	0.45
Weighted F1-Score	0.38

The accuracy indicates that approximately 44.62% of the student outcome predictions in the testing dataset were correctly classified.

The weighted metrics consider the contribution of the different classes according to their support.

5.4.3 Classification Report

The classification report produced the following results.

```
[17] print("CLASSIFICATION REPORT")
print("=" * 60)

print(
    classification_report(
        y_test,
        y_pred,
        zero_division=0
    )
)
```

...

CLASSIFICATION REPORT
=====

```

                precision    recall  f1-score   support

   High           0.39        0.38        0.38        1171
   Low            0.80        0.80        0.80        3077
   Medium         0.43        0.43        0.43        1102

 accuracy              0.63              5350
 macro avg           0.54              5350
 weighted avg        0.63              5350

```

Figure 5.6: Random Forest classification report

Table 5.4: Classification Performance

Class	Precision	Recall	F1-Score	Support
Distinction	0.36	0.01	0.01	605
Fail	0.41	0.06	0.11	1,411
Pass	0.44	0.73	0.55	2,472
Withdrawn	0.46	0.49	0.48	2,031
Accuracy			0.45	6,519
Macro Average	0.42	0.32	0.29	6,519
Weighted Average	0.43	0.45	0.38	6,519

The results show that the model performed differently across the four outcome classes.

The strongest recall was obtained for the Pass category, with a recall of 0.73. This indicates that the model was relatively more successful in identifying students belonging to this category.

The model showed weaker recall for the Distinction and Fail categories. This may be influenced by class imbalance, feature limitations or similarities between academic patterns in different outcome groups.

5.4.4 Results Analysis

The overall accuracy of approximately 44.62% demonstrates that the model is capable of learning some patterns from the selected features, but its predictive performance remains limited.

The results should therefore be interpreted carefully.

Several factors may have affected the performance:

Limited Number of Features

The implemented model used ten features from studentInfo.csv, consisting of eight categorical features (code module, code presentation, gender, region, highest education, IMD band, age band and disability) and two numerical features (number of previous attempts and studied credits).

Student academic performance is influenced by many other factors that were not included in the final model.

Class Distribution

The different outcome categories do not contain identical numbers of students. This may influence the ability of the model to predict minority classes.

Feature Representation

More detailed information relating to learning engagement and Virtual Learning Environment interactions may improve future predictive performance.

Educational Complexity

Student performance is influenced by academic, behavioural and contextual factors. A small set of numerical features may not fully represent this complexity.

5.4.5 Confusion Matrix

A confusion matrix is useful for examining how predictions are distributed across the different classes.

The diagonal values represent correct predictions, while the off-diagonal values represent classification errors.

The implemented Machine Learning evaluation workflow supports the generation of a confusion matrix.

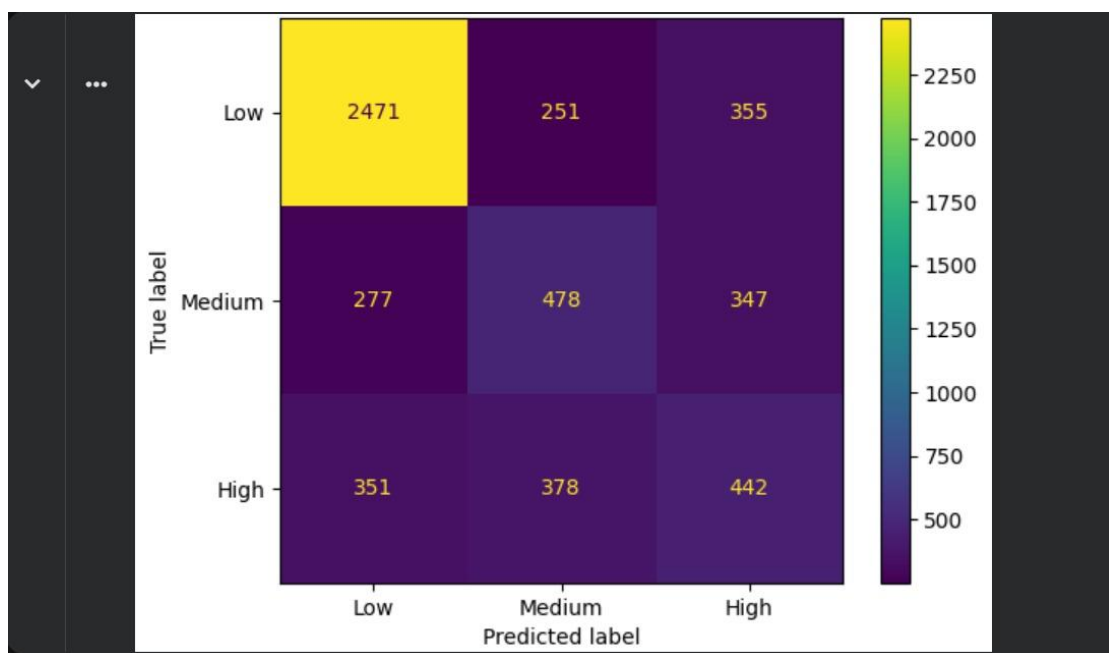


Figure 5.7: Confusion matrix of the Random Forest classifier

5.4.6 Feature Importance

Random Forest can calculate feature importance values to indicate the relative contribution of input features to the model.

The evaluated features include:

1. Average assessment score
2. Assessment count
3. Studied credits

4. Number of previous attempts

Feature importance analysis can help identify which variables contribute most strongly to the model's predictions.

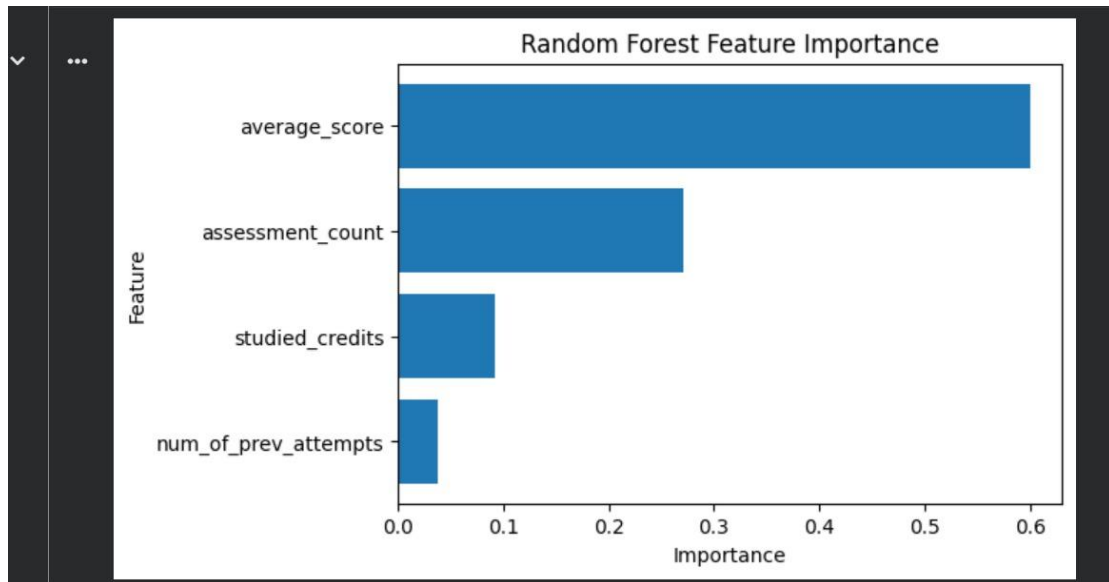


Figure 5.8: Random Forest feature importance

CHAPTER 6: DISCUSSION

6.1 Achievement of Objectives

Objective 1: Develop an AI-based Student Performance Prediction System

This objective was achieved through the development of an integrated system consisting of a Machine Learning model, knowledge base, AI workflow and application prototype.

Objective 2: Prepare and Process Student Data

This objective was achieved by obtaining and processing relevant information from the OULAD dataset.

Relevant CSV files were examined and used to prepare the student data for the Machine Learning workflow.

Objective 3: Develop and Evaluate a Machine Learning Model

A Random Forest Classifier was implemented and trained using processed student data.

The model was successfully trained and saved as risk_model.pkl.

The final execution produced an accuracy of approximately 44.62%.

Although the performance is moderate, the objective of implementing and evaluating a Machine Learning classification model was achieved.

Objective 4: Develop a Knowledge Base and Risk Workflow

This objective was achieved through the creation of a structured academic intervention knowledge base.

The workflow can:

- Identify academic risk factors.
- Retrieve risk-level guidance.
- Retrieve risk-factor-specific recommendations.

Objective 5: Integrate Generative AI

The system integrates a Generative AI component to generate concise academic intervention recommendations.

The AI workflow uses structured prompts, few-shot examples and knowledge-base information.

Objective 6: Develop an Application Prototype

A prototype application was developed to demonstrate the system workflow and allow interaction with the developed components.

Input validation and result display functionality were tested during development.

6.2 Challenges Encountered

Several challenges were encountered during the development of the project.

6.2.1 Dataset Complexity

OULAD contains multiple CSV files that represent different aspects of student activity.

Combining and processing these files requires careful use of identifiers and data relationships.

6.2.2 Model Performance

The final model accuracy was approximately 44.62%, demonstrating that predicting student outcomes is a challenging task.

The limited feature set and class distribution may have affected the results.

6.2.3 Development Environment

The project required the installation and configuration of Python packages.

During implementation, missing dependencies such as Pandas and Scikit-learn had to be installed before the model could be executed successfully.

6.2.4 Code Compatibility

Some code prepared in Google Colab included notebook-specific commands. For example, commands beginning with `!pip` are valid in notebook environments but produce syntax errors when directly executed as a standard Python script.

This required awareness of the differences between Google Colab and local Python execution environments.

6.2.5 API Integration

Integrating Generative AI requires appropriate prompt construction and API configuration.

The project addressed this by structuring the workflow using:

- System prompts
- Few-shot examples
- Student data formatting
- Knowledge-base context

6.3 Limitations of the System

6.3.1 Moderate Model Accuracy

The Machine Learning model achieved an accuracy of approximately 44.62%.

Therefore, the model should not be considered sufficiently accurate for high-stakes academic decisions without further improvement.

6.3.2 Limited Features

Only selected academic features were used in the implemented model.

Additional features may improve prediction performance.

6.3.3 Dataset Context

OULAD represents a specific educational environment and may not directly represent all universities or educational institutions.

6.3.4 Prototype Nature

The application is an academic prototype and does not include a full production-level database or enterprise authentication system.

6.3.5 Rule-Based Risk Thresholds

The intervention workflow uses predefined thresholds for GPA, attendance, missed assignments and participation.

These thresholds may need to be adjusted according to institutional policies or validated using real academic data.

6.3.6 Generative AI Dependence

The quality of AI-generated recommendations depends on:

- Prompt design
- Available context
- API availability
- Generative model behaviour

Therefore, AI recommendations should support rather than replace professional academic judgement.

CHAPTER 7: CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Student Performance Prediction System was developed to explore the application of Artificial Intelligence and Machine Learning in student performance analysis.

The project used the Open University Learning Analytics Dataset to develop a Random Forest classification model for predicting student academic outcomes. The implemented model used selected features including average assessment score, assessment count, studied credits and the number of previous attempts.

The final model was trained on 26,074 students and evaluated using a testing dataset containing 6,519 students. The evaluation produced an accuracy of approximately 44.62%. The results demonstrate that the model was able to identify some patterns within the data, although further improvement is required before the model could be considered highly reliable.

The project also extended beyond Machine Learning prediction by developing an academic intervention workflow. A structured knowledge base was used to identify academic risk levels and risk factors such as low GPA, poor attendance, missed assignments and low participation.

The workflow integrates this information with Generative AI to produce concise and supportive academic intervention recommendations.

Overall, the project demonstrates the potential of integrating:

- Educational data analytics
- Machine Learning
- Rule-based reasoning
- Knowledge representation
- Prompt engineering
- Generative AI

Although the system has limitations, it provides a useful prototype and foundation for future development of more advanced student performance monitoring systems.

7.2 Future Improvements

Several improvements can be made in future work.

7.2.1 Use Additional Features

Future versions can include more information such as:

- Virtual Learning Environment activity
- Frequency of online learning
- Assignment submission patterns
- Registration behaviour
- Historical academic performance

7.2.2 Improve Machine Learning Performance

Additional algorithms could be compared with Random Forest, including:

- XGBoost
- Gradient Boosting
- Support Vector Machines
- Neural Networks

Hyperparameter optimisation could also be applied.

7.2.3 Address Class Imbalance

Future research can investigate methods such as:

- Oversampling
- Undersampling
- SMOTE
- Improved class weighting

These approaches may improve performance for underrepresented classes.

7.2.4 Use Real-Time Data

A future system could be connected to a university learning management system.

This could allow the system to analyse updated student information continuously.

7.2.5 Improve the User Interface

The prototype could be expanded with:

- Student dashboards
- Performance visualisations
- Historical trend analysis
- Advisor management features
- Search and filtering

7.2.6 Database Integration

A future version could use a database such as MySQL or MongoDB to store:

- Student records
- Prediction history
- Intervention history
- Advisor feedback

7.2.7 Human Evaluation

Future work could involve formal testing with:

- Students
- Lecturers
- Academic advisors

This would provide feedback regarding usability and the usefulness of the generated recommendations.

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APPENDIX A: PROJECT SCHEDULE (GANTT CHART)

Project Schedule

Activity	Week 1	Week 2	Week 3	Week 4
Project planning	✓			
GitHub repository setup	✓			
Dataset collection	✓	✓		
Literature review	✓	✓		
Knowledge base preparation		✓		
API integration		✓		
AI workflow development		✓	✓	
Machine Learning development		✓	✓	
Application development			✓	
Functional testing			✓	✓
Performance evaluation			✓	✓
Final report				✓
Presentation preparation				✓
Poster preparation				✓

Figure A.1: Project Gantt Chart

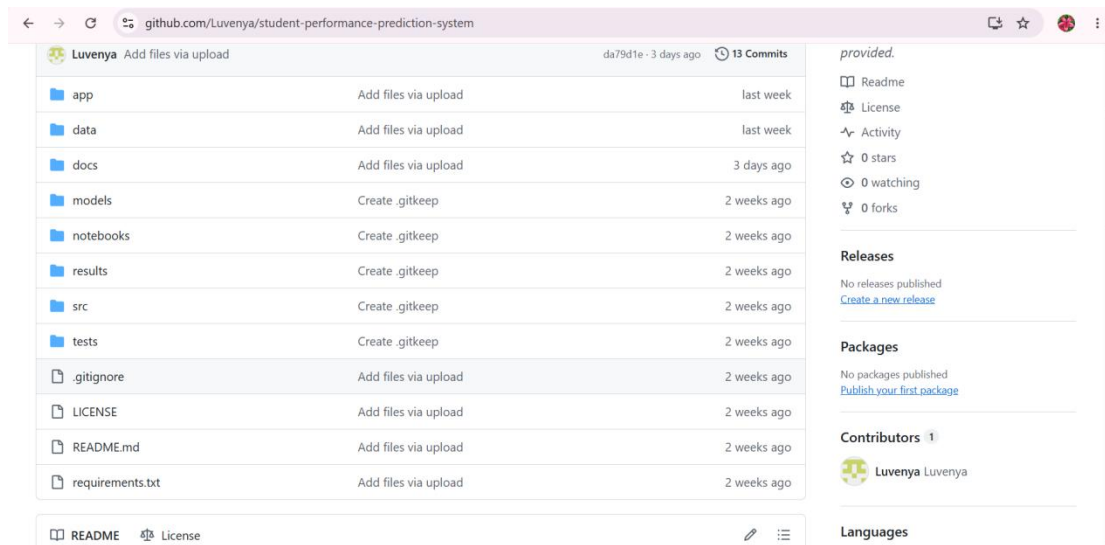
APPENDIX B: SOURCE CODE

The project source code is organised into several modules, including:

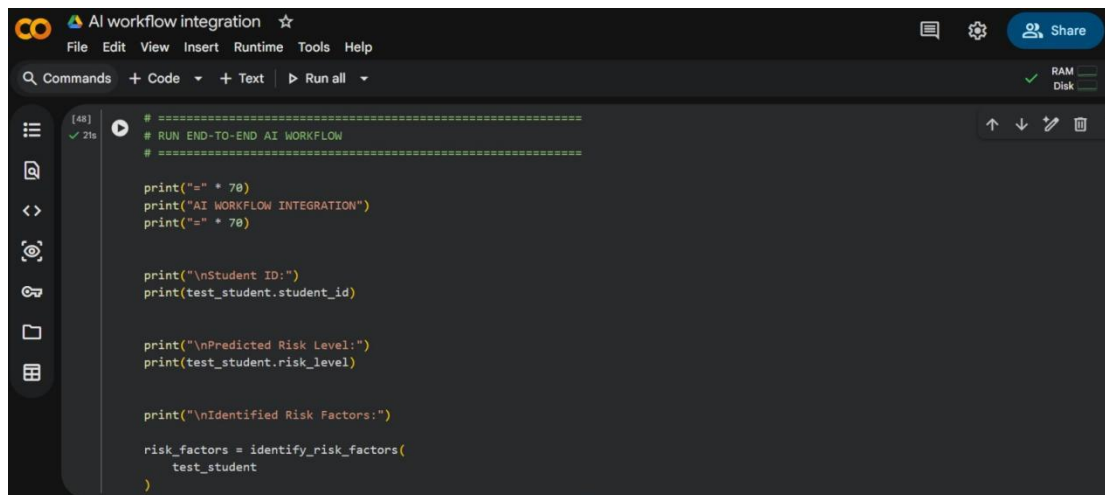
- ai_workflow_integration.py
- api_integration.py
- app.py
- knowledge_base_preparation.py
- llm_service.py
- ml_model_training_and_performance_evaluation.py
- model.py
- prompt_engineering.py
- prototype.py
- train_model.py

The trained Machine Learning model is stored as:

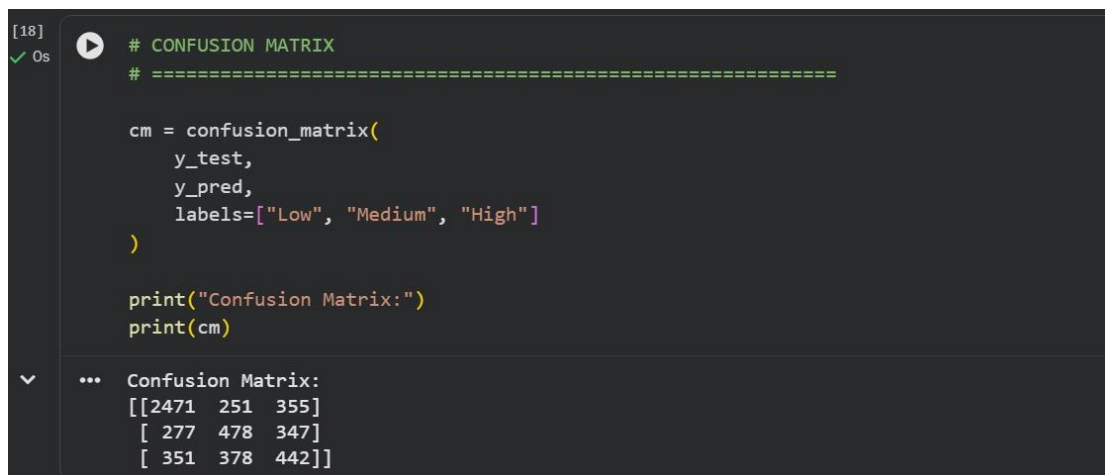
- risk_model.pkl



Github repository



Workflow execution



Confusion matrix code

APPENDIX C: DATASET SAMPLES

The OULAD dataset contains several CSV files.

Example Dataset Files

- studentInfo.csv
- studentRegistration.csv
- courses.csv
- assessments.csv
- studentAssessment.csv
- vle.csv
- studentVle.csv

The screenshot shows a WPS spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I
	code_module	code_presentation	id_student	gender	region	highest_education	imd_band	age_band	num_of
1	AAA	2013J	11391	M	East Anglian Region	HE Qualification	90-100%	55<=	
2	AAA	2013J	28400	F	Scotland	HE Qualification	20-30%	35-55	
3	AAA	2013J	30268	F	North Western Region	A Level or Equivalent	30-40%	35-55	
4	AAA	2013J	31604	F	South East Region	A Level or Equivalent	50-60%	35-55	
5	AAA	2013J	32885	F	West Midlands Region	Lower Than A Level	50-60%	0-35	
6	AAA	2013J	38053	M	Wales	A Level or Equivalent	80-90%	35-55	
7	AAA	2013J	45462	M	Scotland	HE Qualification	30-40%	0-35	
8	AAA	2013J	45642	F	North Western Region	A Level or Equivalent	90-100%	0-35	
9	AAA	2013J	52130	F	East Anglian Region	A Level or Equivalent	70-80%	0-35	
10	AAA	2013J	53025	M	North Region	Post Graduate Qualification	?	55<=	
11	AAA	2013J	57506	M	South Region	Lower Than A Level	70-80%	35-55	
12	AAA	2013J	58873	F	East Anglian Region	A Level or Equivalent	20-30%	0-35	
13	AAA	2013J	59185	M	East Anglian Region	Lower Than A Level	60-70%	35-55	
14	AAA	2013J	62155	F	North Western Region	HE Qualification	50-60%	0-35	
15	AAA	2013J	63400	M	Scotland	Lower Than A Level	40-50%	35-55	
16	AAA	2013J	65002	F	East Anglian Region	A Level or Equivalent	70-80%	0-35	
17	AAA	2013J	70464	F	West Midlands Region	A Level or Equivalent	60-70%	35-55	

studentInfo.csv sample

The screenshot shows a WPS spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K	L	M
	id_assessment	id_student	date_submitted	is_banked	score								
1	1752	11391	18	0	78								
2	1752	28400	22	0	70								
3	1752	31604	17	0	72								
4	1752	32885	26	0	69								
5	1752	38053	19	0	79								
6	1752	45462	20	0	70								
7	1752	45642	18	0	72								
8	1752	52130	19	0	72								
9	1752	53025	9	0	71								
10	1752	57506	18	0	68								
11	1752	58873	19	0	73								
12	1752	59185	18	0	67								
13	1752	62155	17	0	73								
14	1752	63400	19	0	83								
15	1752	65002	17	0	66								
16	1752	70464	19	0	59								
17	1752	71361	19	0	82								

studentAssessment.csv sample

APPENDIX D: USER MANUAL

Step 1: Start the Application

Open the project directory and start the Flask backend by running `python app.py`.

Then open the frontend `index.html` file in a web browser.

Step 2: Enter Student Information

Enter the required student information into the application.

Ensure that:

- Student IDs are numeric where required.
- Required fields are not left invalid.
- Values are entered in the appropriate format.

Step 3: Submit the Information

Click the relevant button to process the student information.

The system validates the input before processing.

Step 4: View the Results

The application displays the relevant results, which may include:

- Student information
- Prediction results
- Summary information
- Identified risk factors
- Academic intervention recommendations

Step 5: Review the Recommendation

The AI-generated recommendation should be reviewed as supportive academic guidance.

The recommendation is designed to be:

- Professional
- Supportive
- Non-judgmental
- Action-oriented
- Concise